

ALEXANDER C. JENKINS

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ABOUT ME

I'm a theoretical physicist, working at the interface between *cosmology*, *astrophysics*, *high-energy physics*, and *quantum matter*. My research looks at new ways of probing the fundamental laws of Nature, whether that's using *gravitational waves* as powerful new astronomical messengers, or using cutting-edge *quantum technologies* to simulate the early Universe.

EMPLOYMENT

Gavin Boyle Fellow in Cosmology — *University of Cambridge* *2024–present*

UKRI Stephen Hawking Fellow *from May 2025*

Research fellow hosted in the [Kavli Institute for Cosmology Cambridge](#) (KICC) and the [Department of Applied Mathematics and Theoretical Physics](#) (DAMTP) | Fellow of [Selwyn College](#)

Postdoctoral Research Fellow — *University College London* *2021–2024*

Led an international, interdisciplinary project to study vacuum decay with quantum analogue experiments and lattice simulations, as part of the [QSimFP Consortium](#) | Member of the [Cosmoparticle Initiative](#)

EDUCATION

PhD in Theoretical Physics — *King's College London* *2017–2021*

Funded by competitive faculty scholarship | [Theoretical Particle Physics and Cosmology Group](#)

'*Cosmology and fundamental physics in the era of GW astronomy*', supervised by [Mairi Sakellariadou](#)

MSci in Astrophysics (Part III) — *University of Cambridge* *2016–2017*

1st class | Ranked 5th in cohort | Elected a Bateman Scholar of Trinity Hall for 'excellent exam results'

Leonid Grishchuk Internship Program — *Cardiff University* *Summer 2016*

Competitive, funded summer research internship in the [Gravitational Physics Group](#)

MA in Natural Sciences (Astrophysics) — *University of Cambridge* *2013–2016*

1st class | Elected a Scholar of Trinity Hall

GRANTS AND FUNDING SECURED (POST-PHD)

- UKRI Stephen Hawking Fellowship (PI, £506k fEC) *2024*
3-year research council fellowship supporting 'visionary scientists working in theoretical physics'
- Gavin Boyle Fellowship, Kavli Institute for Cosmology and Selwyn College, Cambridge *2024*
5-year institutional fellowship
- UKRI Quantum Technologies for Fundamental Physics Grant extension (Co-I, £69k) *2023*
Wrote successful proposal for 6-month funded extension to my QSimFP research

AFFILIATIONS

Scientific collaborations

* Quantum Simulators for Fundamental Physics (QSimFP) Consortium (*2021–present*) * LISA Consortium (*2018–present*) * Einstein Telescope (ET) Observational Science Board (*2021–present*) * LIGO Scientific Collaboration (*2016–2021*)

Professional bodies

* Member, Institute of Physics (MInstP; *2018–present*) * Fellow, Royal Astronomical Society (FRAS; *2020–present*) * Junior Member, European Astronomical Society (*2020–present*)

AWARDS AND ACHIEVEMENTS

- Winner, [Buchalter Cosmology Prize](#) (2nd Prize) | [UCL press release](#) 2023
International award recognising ‘ground-breaking theoretical, observational, or experimental work in cosmology that has the potential to produce a breakthrough advance in our understanding’
- Honorable Mention in the [GWIC-Braccini Thesis Prize](#) competition 2022
Nominated for three other thesis prizes
- Best Student Talk Prize at [BritGrav 21](#), sponsored by IoP Publishing 2021
Corresponding paper published in *Classical and Quantum Gravity* as an invited submission
- [King’s Education Award](#) for ‘extraordinary contributions’ to teaching 2020
Also nominated as a ‘Rising Star’ in the 2019 awards (only PhD student nominee in Physics)
- Bateman Scholar of Trinity Hall (Cambridge), recognising ‘excellent exam results’ 2017

RESPONSIBILITIES

Organisation of scientific meetings and seminars

- Lead organiser, [Kavli Focus Meeting on Gravitational Waves](#) 2025
Initiated and led the organisation of a local workshop bringing together an interdisciplinary mix of researchers across three different Cambridge departments, with funding secured from KICC
- Co-organiser, workshop on ‘Gravitational Waves and Dark Matter in Orbital Dynamics’ 2025
SOC member for workshop in Barcelona, funding secured from the [Gravity Theory Trust](#) (\$18k)
- Lead organiser, *UCL Cosmology/Extragalactic Seminars* 2022–2023
Developed an ambitious program of in-person talks with speakers from across the UK and abroad, from what had previously been an online-only event due to COVID-19; secured and managed small grant (£1k) for speaker expenses and group lunches to encourage student participation
- Co-organiser, [London Cosmology Discussion Meetings \(LCDM\)](#) 2018–2021
Coordinated between five institutions to organise meetings at the Royal Astronomical Society on ‘Dark Matter in Cosmology’, ‘Neutrinos in Cosmology’, and ‘Cosmological Probes of New Physics’
- Organiser, *Theoretical Particle Physics and Cosmology (TPPC) Journal Club* 2021
- Organiser, *TPPC Gravity Meetings* 2018–2020
Initiated a regular series of meetings with internal and external speakers on gravitational physics

Refereeing and review work

- Expert referee for one ERC Consolidator Grant, one NSF Research Grant, 2019–present
one UKRI Stephen Hawking Fellowship, and one Royal Society Dorothy Hodgkin Fellowship
- Referee for 29 articles in *Physical Review Letters (PRL)*, *Nature Astronomy*, 2020–present
Physical Review D (PRD), *Journal of Cosmology and Astroparticle Physics (JCAP)*,
European Physical Journal C (EPJC), *The Astronomical Journal*, and *Universe*

Committees and institutional service

- Elector, Selwyn College Mastership Election 2024–2025
Contributed to interviews and selection process for the new head of my Cambridge college
- Selwyn College Governing Body and IT and Data Committee 2024–present
- Student representative, KCL Physics Department Research Committee 2019–2021

TEACHING AND SUPERVISION

University of Cambridge

2024–present

- Delivered three guest lectures and led small-group ‘supervision’ teaching for 3rd-year *Relativity* (Part II Physics/Astrophysics); student feedback highlighted my ‘very clear and engaging lecturing’

University College London

2021–2024

- Lead supervisor of research projects for two masters students: Phoebe Routh ([distinction](#)) and David Moody ([distinction](#) and awarded departmental prize)
- Postgraduate Teaching Assistant for 3rd-year *Physical Cosmology*: developed problem sets, delivered problem-solving tutorials, and contributed to marking of final exams

King’s College London

2017–2021

- Winner of a 2020 [King’s Education Award](#), recognising ‘extraordinary contributions’ to teaching
- ‘Rising Star’ nominee in the 2019 King’s Education Awards (only PhD student nominee in Physics)
- Co-wrote lecture notes for 3rd-year *General Relativity and Cosmology*
- Examples class demonstrator for numerous courses, including 4th-year *Astroparticle Cosmology*, 3rd-year *General Relativity and Cosmology*, 2nd-year *Astrophysics*, 1st-year *Mathematics for Physicists*, ...

SELECTED PUBLIC ENGAGEMENT

- Contributed to the art/science exhibition ‘*Cosmic Titans*’ at Lakeside Arts, Nottingham 2025
Interacted with artist Conrad Shawcross RA to influence his piece ‘*The Blind Proliferation*’; gave a [video interview](#) exploring the science behind Conrad’s work, featured in the exhibition and online
- My research and simulations featured in the documentary ‘*Do we live in a multiverse?*’ 2024
Aired on [French](#) and [German](#) TV and more than 5 million combined views on YouTube
- Led maths outreach activities at a local school as part of the [Cambridge NRICH project](#) 2024
- Invited speaker for the [Cambridge Astronomical Association](#) 2024
- YouTube video interview on ‘*Early Universe Cosmology in the Lab*’ 2023
- Outreach talk for alumni of UCL’s ‘[Introduction to Astronomy](#)’ course, 2023
aimed at amateur astronomers and members of the public
- Participated in five interviews for media pieces on my paper 2022
‘*Bridging the μHz gap in the gravitational-wave landscape with binary resonance*’
- Maths and physics tutor at Open Tutors London 2017–2020
Co-initiated free tutoring program for University of London students from under-represented groups
- Helped run an interactive exhibit on Dark Matter at [Science Gallery London](#) 2019
- Local organiser, [Pint of Science Festival](#) 2018

SOFTWARE AND NUMERICS

- Author of Fortran lattice field theory code `lattice-fvd` and Python code [gw-resonance](#)
- Experience with advanced numerical methods including, e.g., Fourier and Chebyshev pseudospectral methods and symplectic integration
- Extensive experience in Unix environments (Ubuntu/MacOS), including in HPC settings; extensive Python experience (object-oriented programming; data handling and visualisation; [Jupyter](#), [NumPy](#), [SciPy](#), [h5py](#), [Astropy](#), [healpy](#), [sympy](#), ...); other languages and software include Fortran, C++, Mathematica, Git, MATLAB, SageMath, SQL, $\text{\LaTeX}$ (including [TikZ](#)), ...

SELECTED PUBLICATIONS

Citation statistics as of 17 March 2025 (data from [INSPIRE-HEP](#)):

Lead-author only	16 papers, 637 citations, h -index = 11
Non-LIGO only	29 papers, 2,656 citations, h -index = 19
All publications	104 papers, 33,515 citations, h -index = 64

Lead-author papers (author list is ordered alphabetically in some cases)

- L1. **ACJ**, I. G. Moss, T. P. Billam, Z. Hadzibabic, H. V. Peiris, and A. Pontzen, *Generalized cold-atom analogues for vacuum decay* (2024), *Phys. Rev. A* **110**, L031301, [arXiv:2311.02156 \[cond-mat.quant-gas\]](#) | [Letter](#)
- L2. **ACJ**, J. Braden, H. V. Peiris, A. Pontzen, M. C. Johnson, and S. Weinfurtner, *Analog vacuum decay from vacuum initial conditions* (2024), *Phys. Rev. D* **109**, 023506, [arXiv:2307.02549 \[cond-mat.quant-gas\]](#) | [Editor's Suggestion](#)
- L3. A. K.-W. Chung, **ACJ**, J. D. Romano, and M. Sakellariadou, *Targeted search for the kinematic dipole of the gravitational-wave background* (2022), *Phys. Rev. D* **106**, 082005, [arXiv:2208.01330 \[gr-qc\]](#)
- L4. M. R. Mosbech, **ACJ**, S. Bose, C. Boehm, M. Sakellariadou, and Y. Y. Y. Wong, *Gravitational-wave event rates as a new probe for dark matter microphysics* (2023), *Phys. Rev. D* **108**, 043512, [arXiv:2207.14126 \[astro-ph.CO\]](#)
Co-lead author with Markus Mosbech; I developed the core idea and led $\sim 50\%$ of the analysis
[Featured](#) in a [Royal Astronomical Society press release](#) at the 2023 National Astronomy Meeting
- L5. **ACJ**, *Cosmology and Fundamental Physics in the Era of Gravitational-Wave Astronomy* (2022, PhD thesis), [arXiv:2202.05105 \[gr-qc\]](#)
- L6. D. Blas and **ACJ**, *Bridging the μHz gap in the gravitational-wave landscape with binary resonance* (2022), *Phys. Rev. Lett.* **128**, 101103, [arXiv:2107.04601 \[astro-ph.CO\]](#)
Awarded a [Buchalter Cosmology Prize](#) (2nd Prize), recognising 'potential for remarkable impact'
Altmetric [attention score](#) of 395, in the top 0.3% of all publications ever tracked by Altmetric
[Featured](#) in the *Daily Express*, *Physics* magazine, *Big Think*, *SYFY wire*, and 40+ other outlets
- L7. D. Blas and **ACJ**, *Detecting stochastic gravitational waves with binary resonance* (2022), *Phys. Rev. D* **105**, 064021, [arXiv:2107.04063 \[gr-qc\]](#)
- L8. **ACJ** and M. Sakellariadou, *Nonlinear gravitational-wave memory from cusps and kinks on cosmic strings* (2021), *Class. Quant. Grav.* **38**, 165004, [arXiv:2102.12487 \[gr-qc\]](#)
[Invited submission](#) to CQG as winner of the Best Student Talk Prize at BritGrav 21
- L9. **ACJ** and M. Sakellariadou, *Primordial black holes from cusp collapse on cosmic strings* (2020), [arXiv:2006.16249 \[astro-ph.CO\]](#)
- L10. **ACJ**, J. D. Romano, and M. Sakellariadou, *Estimating the angular power spectrum of the gravitational-wave background in the presence of shot noise* (2019), *Phys. Rev. D* **100**, 083501, [arXiv:1907.06642 \[astro-ph.CO\]](#)
- L11. **ACJ** and M. Sakellariadou, *Shot noise in the astrophysical gravitational-wave background* (2019), *Phys. Rev. D* **100**, 063508, [arXiv:1902.07719 \[astro-ph.CO\]](#)
- L12. **ACJ**, R. O'Shaughnessy, M. Sakellariadou, and D. Wysocki, *Anisotropies in the astrophysical gravitational-wave background: The impact of black hole distributions* (2019), *Phys. Rev. Lett.* **122**, 111101, [arXiv:1810.13435 \[astro-ph.CO\]](#)
- L13. **ACJ**, A. G. A. Pithis, and M. Sakellariadou, *Can we detect quantum gravity with compact binary inspirals?* (2018), *Phys. Rev. D* **98**, 104032, [arXiv:1809.06275 \[gr-qc\]](#)

- L14. **ACJ**, M. Sakellariadou, T. Regimbau, and E. Slezak, *Anisotropies in the astrophysical gravitational-wave background: Predictions for the detection of compact binaries by LIGO and Virgo* (2018), Phys. Rev. D **98**, 063501, arXiv:1806.01718 [astro-ph.CO]
- L15. **ACJ** and M. Sakellariadou, *Anisotropies in the stochastic gravitational-wave background: Formalism and the cosmic string case* (2018), Phys. Rev. D **98**, 063509, arXiv:1802.06046 [astro-ph.CO]
Featured in PRD's 'kaleidoscope' for Sep 2018

Other selected papers (with summary of my main contributions)

- O1. L. Zwick, D. Soyuer, D. J. D'Orazio, D. O'Neill, A. Derdzinski, P. Saha, D. Blas, **ACJ**, L. Z. Kelley, *Bridging the micro-Hz gravitational wave gap via Doppler tracking with the Uranus Orbiter and Probe Mission: Massive black hole binaries, early universe signals and ultra-light dark matter* (2024), arXiv:2406.02306 [astro-ph.HE] | Led sensitivity analysis for early-Universe signals
- O2. N. Kouvatsos, **ACJ**, A. I. Renzini, J. D. Romano, M. Sakellariadou, *Unbiased estimation of gravitational-wave anisotropies from noisy data* (2023), arXiv:2312.09110 [astro-ph.CO] | Proposed new analysis method and led theoretical work; informal supervision of PhD student (N. Kouvatsos)
- O3. M. Branchesi *et al.*, *Science with the Einstein Telescope: a comparison of different designs* (2023), JCAP **07**, 068, arXiv:2303.15923 [gr-qc] | Contributed to stochastic background sensitivity analysis for different Einstein Telescope configurations, guiding further development of the proposal
- O4. S. Gasparrotto, R. Vicente, D. Blas, **ACJ**, and E. Barausse, *Can gravitational-wave memory help constrain binary black-hole parameters? A LISA case study* (2023), Phys. Rev. D **107**, 124033, arXiv:2301.13228 [gr-qc] | Helped define project and methodologies; informal supervision of PhD student (S. Gasparrotto)
- O5. P. Auclair *et al.* (LISA Cosmology Working Group), *Cosmology with the Laser Interferometer Space Antenna* (2022), Living Rev. Rel. **26**, 5, arXiv:2204.05434 [astro-ph.CO] | LISA white paper; contributed to section on cosmic strings, led analysis of related gravitational-wave anisotropies
- O6. A. I. Renzini, B. Goncharov, **ACJ**, and P. M. Meyers, *Stochastic Gravitational-Wave Backgrounds: Current Detection Efforts and Future Prospects* (2022), Galaxies **10**, 34, arXiv:2202.00178 [gr-qc] | Invited review article; major contributions to sections on gravitational-wave theory and sources
- O7. N. Bartolo *et al.* (LISA Cosmology Working Group), *Probing Anisotropies of the Stochastic Gravitational Wave Background with LISA* (2022), JCAP **11**, 009, arXiv:2201.08782 [astro-ph.CO] | LISA review paper; coordinator for 'topological defects' section, with further contributions to 'astrophysical sources' section
- O8. P. Auclair, J. J. Blanco-Pillado, D. G. Figueroa, **ACJ**, M. Lewicki, M. Sakellariadou, S. Sanidas, L. Sousa, D. A. Steer, J. M. Wachter, and S. Kuroyanagi (LISA Cosmology Working Group), *Probing the gravitational wave background from cosmic strings with LISA* (2019), JCAP **04**, 034, arXiv:1909.00819 [astro-ph.CO] | LISA review paper; significant writing contributions throughout
- O9. B. P. Abbott *et al.* (LIGO, Virgo), *Directional limits on persistent gravitational waves using data from Advanced LIGO's first two observing runs* (2019), Phys. Rev. D **100**, 062001, arXiv:1903.08844 [gr-qc] | Led interpretation of observational results in the context of cosmological and astrophysical source models, wrote corresponding section
- O10. B. P. Abbott *et al.* (LIGO, Virgo), *Search for the isotropic stochastic background using data from Advanced LIGO's second observing run* (2019), Phys. Rev. D **100**, 061101, arXiv:1903.02886 [gr-qc] | Rapid communication | Led and wrote section on implications for cosmic string models

SELECTED INVITED TALKS

Total of 45 invited talks across three continents

- **Numerical simulations of early Universe sources of gravitational waves** *Aug 2025*
Nordita, Stockholm
- **Multiverse from Home** *Jun 2025*
Online
- **WE-Heraeus Seminar: New Windows on the Universe** *May 2025*
Kitzbühel, Austria
- **QSimFP Community Workshop** *Mar 2025*
University of Nottingham
- **Early Universe from Home** *Feb 2025*
Online
- **Theoretical Particle Physics Seminar** *Feb 2025*
University of Sussex
- **Joint seminar, LMU Cosmology and MPI Quantum Optics** *Jan 2025*
Ludwig Maximilian University of Munich
- **General Relativity Seminar** *Dec 2024*
DAMTP, University of Cambridge
- **Cosmology Journal Club** *Nov 2024*
University of Padova (online)
- **Particle Cosmology Seminar** *Nov 2024*
University of Nottingham
- **Cosmology/Extragalactic Seminar** *Oct 2024*
University College London
- **Cambridge-LMU Meeting** *Oct 2024*
Kavli Institute for Cosmology, University of Cambridge
- **GEMMA2 Workshop** *Sep 2024*
Sapienza University of Rome
- **Majorana-Raychaudhuri Seminar** *Sep 2024*
Kolkata/Salerno (online)
- **Cold atoms and molecules for fundamental physics** *Jul 2024*
Cambridge
- **4th EuCAPT Symposium** *May 2024*
CERN, Geneva
- **British Applied Mathematics Colloquium (BAMC)** *Apr 2024*
Newcastle University
- **Cosmology Lunch Seminar** *Feb 2024*
DAMTP, University of Cambridge
- **Gravitational-Wave Group Meeting** *Jan 2024*
Institute of Cosmology and Gravitation (ICG), University of Portsmouth

- **Oberthaler Group Seminar** *Nov 2023*
Kirchhoff Institute for Physics (KIP), Heidelberg
- **Cosmology from Home** *Jul 2023*
Online
- **Quantum Simulators for Fundamental Physics Workshop** *Jun 2023*
Perimeter Institute for Theoretical Physics, Waterloo, Canada
- **Astrophysics Seminar** *May 2023*
University of Leicester
- **Cosmology Seminar** *May 2023*
Beecroft Institute, Oxford
- **Theory Group Seminar** *May 2023*
Astroparticle and Cosmology Laboratory (APC), Paris
- **Cosmology and Relativity Seminar** *Apr 2023*
Queen Mary University of London
- **London Gravity Meeting** *Mar 2023*
Royal Society, London
- **‘Dark Matters’ Workshop** *Nov 2022*
Université Libre de Bruxelles (ULB)
- **London-Oldenburg Relativity Seminar** *Nov 2022*
University College London/University of Oldenburg (online)
- **ICTP-AP Seminar** *Sep 2022*
International Centre for Theoretical Physics, Asia-Pacific (online)
- **Quantum Simulators for Fundamental Physics Workshop** *Sep 2022*
Science Gallery London
- **‘Gravitational-Wave Orchestra’ Workshop** *Sep 2022*
Université Catholique de Louvain, Belgium
- **Theory Group Seminar** *May 2022*
Institute of High-Energy Physics (IFAE), Barcelona
- **Quantum Technology Seminar** *May 2022*
London Centre for Nanotechnology, University College London
- **Theory Group Seminar** *Oct 2021*
Astroparticle and Cosmology Laboratory (APC), Paris
- **Ibarra Group Seminar** *Jul 2021*
Technical University of Munich (online)
- **Gravitational Wave Probes of Physics Beyond the Standard Model** *Jul 2021*
University of Warsaw (online)
- **Theoretical Cosmology Seminar** *May 2020*
Institute of Cosmology and Gravitation (ICG), University of Portsmouth (online)
- **Cosmology Seminar** *May 2020*
Beecroft Institute, University of Oxford (online)
- **Gravitational Wave Probes of Fundamental Physics** *Nov 2019*
EuCAPT workshop, Amsterdam